

Table 1. List of Included Primary Studies

ID	Year	Title	Venue	Reference
1	2021	Unit Test Case Generation with Transformers and Focal Context	arXiv	[1]
2	2022	Generating Accurate Assert Statements for Unit Test Cases using Pre-trained Transformers	AST	[2]
3	2022	TOGA: A Neural Method for Test Oracle Generation	ICSE	[3]
4	2023	An Initial Investigation of ChatGPT Unit Test Generation Capability	SAST	[4]
5	2023	CAT-LM Training Language Models on Aligned Code And Tests	ASE	[5]
6	2023	CodaMosa: Escaping Coverage Plateaus in Test Generation with Pre-trained Large Language Models	ICSE	[6]
7	2023	CodeT: Code Generation with Generated Tests	ICLR	[7]
8	2023	Exploring the Capability of ChatGPT in Test Generation	QRS	[8]
9	2023	Identify and Update Test Cases When Production Code Changes: A Transformer-Based Approach	ASE	[9]
10	2023	Large Language Models are Few-shot Testers: Exploring LLM-based General Bug Reproduction	ICSE	[10]
11	2023	Learning Deep Semantics for Test Completion	ICSE	[11]
12	2023	Neural-Based Test Oracle Generation: A Large-Scale Evaluation and Lessons Learned	FSE	[12]
13	2023	Nuances are the Key: Unlocking ChatGPT to Find Failure-Inducing Tests with Differential Prompting	ASE	[13]
14	2023	Prompting Code Interpreter to Write Better Unit Tests on Quixbugs Functions	arXiv	[14]
15	2023	Retrieval-Based Prompt Selection for Code-Related Few-Shot Learning	ICSE	[15]
16	2023	Towards More Realistic Evaluation for Neural Test Oracle Generation	ISSTA	[16]
17	2023	Using Transfer Learning for Code-Related Tasks	TSE	[17]
18	2024	A3Test: Assertion-Augmented Automated Test case generation	IST	[18]
19	2024	An Empirical Evaluation of Using Large Language Models for Automated Unit Test Generation	TSE	[19]
20	2024	An Empirical Study on Focal Methods in Deep-Learning-Based Approaches for Assertion Generation	FSE	[20]
21	2024	An LLM-based Readability Measurement for Unit Tests' Context-aware Inputs	arXiv	[21]
22	2024	Assessing Evaluation Metrics for Neural Test Oracle Generation	TSE	[22]
23	2024	Automated Unit Test Improvement using Large Language Models at Meta	FSE	[23]
24	2024	Automatic Generation of Test Cases based on Bug Reports: a Feasibility Study with Large Language Models	ICSE	[24]
25	2024	Chat-like Asserts Prediction with the Support of Large Language Model	arXiv	[25]
26	2024	ChatGPT vs SBST: A Comparative Assessment of Unit Test Suite Generation	TSE	[26]
27	2024	ChatUniTest: A Framework for LLM-Based Test Generation	FSE	[27]
28	2024	Code-Aware Prompting: A Study of Coverage-Guided Test Generation in Regression Setting using LLM	FSE	[28]
29	2024	CPP-UT-Bench: Can LLMs Write Complex Unit Tests in C++?	NeurIPS	[29]
30	2024	Deep Multiple Assertions Generation	FORGE	[30]
31	2024	Design choices made by LLM-based test generators prevent them from finding bugs	arXiv	[31]
32	2024	Do LLMs Generate Test Oracles That Capture the Actual or the Expected Program Behaviour?	arXiv	[32]
33	2024	Domain Adaptation for Code Model-Based Unit Test Case Generation	ISSTA	[33]
34	2024	Effective Test Generation using Pre-trained Large Language Models and Mutation Testing	IST	[34]

Continued on next page

ID	Year	Title	Venue	Reference
35	2024	Evaluating and Improving ChatGPT for Unit Test Generation	FSE	[35]
36	2024	Evaluating Large Language Models in Detecting Test Smells	SBES	[36]
37	2024	Evaluation of Large Language Models for Unit Test Generation	ASYU	[37]
38	2024	Fix the Tests: Augmenting LLMs to Repair Test Cases with Static Collector and Neural Reranker	ISSRE	[38]
39	2024	FlakyFix: Using Large Language Models for Predicting Flaky Test Fix Categories and Test Code Repair	TSE	[39]
40	2024	Harnessing the Power of LLMs: Automating Unit Test Generation for High-Performance Computing	arXiv	[40]
41	2024	HITS: High-coverage LLM-based Unit Test Generation via Method Slicing	ASE	[41]
42	2024	Large Language Models as Test Case Generators: Performance Evaluation and Enhancement	arXiv	[42]
43	2024	LLM-Based Test-Driven Interactive Code Generation: User Study and Empirical Evaluation	TSE	[43]
44	2024	LLM4VV: Developing LLM-Driven Testsuite for Compiler Validation	FGCS	[44]
45	2024	LTM: Scalable and Black-Box Similarity-Based Test Suite Minimization Based on Language Models	TSE	[45]
46	2024	Method-Level Test-to-Code Traceability Link Construction by Semantic Correlation Learning	TSE	[46]
47	2024	On the Evaluation of Large Language Models in Unit Test Generation	ASE	[47]
48	2024	Optimizing Search-Based Unit Test Generation with Large Language Models: An Empirical Study	Internetware	[48]
49	2024	Parameter-Efficient Fine-Tuning of Large Language Models for Unit Test Generation: An Empirical Study	arXiv	[49]
50	2024	TestART: Improving LLM-based Unit Test via Co-evolution of Automated Generation and Repair Iteration	arXiv	[50]
51	2024	TestSpark: IntelliJ IDEA's Ultimate Test Generation Companion	ICSE	[51]
52	2024	The Program Testing Ability of Large Language Models for Code	EMNLP	[52]
53	2024	Towards Understanding the Effectiveness of Large Language Models on Directed Test Input Generation	ASE	[53]
54	2024	Transducer Tuning: Efficient Model Adaptation for Software Tasks Using Code Property Graphs	arXiv	[54]
55	2024	Unit Test Generation using Generative AI : A Comparative Performance Analysis of Autogeneration Tools	LLM4Code	[55]
56	2024	Unit Test Generation using Large Language Models for Unity Game Development	FaSE4Games	[56]
57	2024	UniTSyn: A Large-Scale Dataset Capable of Enhancing the Prowess of Large Language Models for Program Testing	ISSTA	[57]
58	2024	Using GitHub Copilot for Test Generation in Python: An Empirical Study	AST	[58]
59	2024	Using Large Language Models to Generate JUnit Tests: An Empirical Study	EASE	[59]
60	2025	A Large-Scale Empirical Study on Fine-Tuning Large Language Models for Unit Testing	ISSTA	[60]
61	2025	AgentTester: An LLM-Based Tool for Unit Test Generation with Automatically Generated Prompts	ICIC	[61]
62	2025	AssertT5: Test Assertion Generation Using a Fine-Tuned Code Language Model	AST	[62]
63	2025	AssertFix: Empowering Automated Assertion Fix via Large Language Models	arXiv	[63]
64	2025	Assertion Messages with Large Language Models (LLMs) for Code	EASE	[64]
65	2025	AssertionBench: A Benchmark to Evaluate Large-Language Models for Assertion Generation	NAACL	[65]

Continued on next page

ID	Year	Title	Venue	Reference
66	2025	ASTER: Natural and Multi-Language Unit Test Generation with LLMs	ICSE	[66]
67	2025	AugmenTest: Enhancing Tests with LLM-Driven Oracles	ICST	[67]
68	2025	Automated Generation of Issue-Reproducing Tests by Combining LLMs and Search-Based Testing	ASE	[68]
69	2025	Automated Test Case Repair Using Language Models	TSE	[69]
70	2025	Automated Unit Test Refactoring	FSE	[70]
71	2025	Automating Autograding: Large Language Models as Test Suite Generators for Introductory Programming	JCAL	[71]
72	2025	CasModaTest: A Cascaded and Model-agnostic Self-directed Framework for Unit Test Generation	ISSRE	[72]
73	2025	ChatAssert: LLM-Based Test Oracle Generation With External Tools Assistance	TSE	[73]
74	2025	Clarifying Semantics of In-Context Examples for Unit Test Generation	ASE	[74]
75	2025	Co-Evolving LLM Coder and Unit Tester via Reinforcement Learning	NeurIPS	[75]
76	2025	Comprehend, Imitate, and then Update: Unleashing the Power of LLMs in Test Suite Evolution	ASE	[76]
77	2025	COTE: Predicting Code-to-Test Co-Evolution by Integrating Link Analysis and Pre-Trained Language Model Techniques	TSE	[77]
78	2025	CoverUp: Effective High Coverage Test Generation for Python	FSE	[78]
79	2025	CubeTesterAI: Automated JUnit Test Generation Using the LLaMA Model	ICST	[79]
80	2025	DeCon: Detecting Incorrect Assertions via Postconditions Generated by a Large Language Model	arXiv	[80]
81	2025	DiffuTester: Accelerating Unit Test Generation for Diffusion LLMs via Mining Structural Pattern	arXiv	[81]
82	2025	Do LLMs Generate Useful Test Oracles? An Empirical Study with an Unbiased Dataset	ASE	[82]
83	2025	Doc2OracLL: Investigating the Impact of Documentation on LLM-Based Test Oracle Generation	FSE	[83]
84	2025	Dynamic Scaling of Unit Tests for Code Reward Modeling	ACL	[84]
85	2025	Enhancing Large Language Models for Text-to-Testcase Generation	JSS	[85]
86	2025	Enhancing LLM's Ability to Generate More Repository-Aware Unit Tests Through Precise Context Injection	ASE	[86]
87	2025	Enriching automatic test case generation by extracting relevant test inputs from bug reports	EMSE	[87]
88	2025	exLong: Generating Exceptional Behavior Tests with Large Language Models	ICSE	[88]
89	2025	Exploring Automated Assertion Generation via Large Language Models	TOSEM	[89]
90	2025	FlakyGuard: Automatically Fixing Flaky Tests at Industry Scale	ASE	[90]
91	2025	FuzzAug: Data Augmentation by Coverage-guided Fuzzing for Neural Test Generation	EMNLP	[91]
92	2025	Improving Deep Assertion Generation via Fine-Tuning Retrieval-Augmented Pre-trained Language Models	TOSEM	[92]
93	2025	Improving Retrieval-Augmented Deep Assertion Generation via Joint Training	TSE	[93]
94	2025	Improving the Readability of Automatically Generated Tests Using Large Language Models	ICST	[94]
95	2025	Learning to Generate Unit Tests for Automated Debugging	COLM	[95]
96	2025	Less Is More: On the Importance of Data Quality for Unit Test Generation	FSE	[96]
97	2025	Leveraging Large Language Models for Enhancing the Understandability of Generated Unit Tests	ICSE	[97]
98	2025	LLM Based Input Space Partitioning Testing for Library APIs	ICSE	[98]

Continued on next page

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99	2025	LLM-enhanced evolutionary test generation for untyped languages	AUSE	[99]
100	2025	LLM-Powered Test Case Generation for Detecting Bugs in Plausible Programs	ACL	[100]
101	2025	LLMs for Automated Unit Test Generation and Assessment in Java: The AgoneTest Framework	ASE	[101]
102	2025	Multi-Agent Assisted Automatic Test Generation for Java JSON Libraries	APSEC	[102]
103	2025	Mutation-Guided LLM-based Test Generation at Meta	FSE	[103]
104	2025	Navigating the Labyrinth: Path-Sensitive Unit Test Generation with Large Language Models	ASE	[104]
105	2025	Nexus: Execution-Grounded Multi-Agent Test Oracle Synthesis	arXiv	[105]
106	2025	NIODebugger: A Novel Approach to Repair Non-Idempotent-Outcome Tests with LLM-Based Agent	ICSE	[106]
107	2025	Otter: Generating Tests from Issues to Validate SWE Patches	ICML	[107]
108	2025	PALM: Synergizing Program Analysis and LLMs to Enhance Rust Unit Test Coverage	ASE	[108]
109	2025	Pytester: Deep reinforcement learning for text-to-testcase generation	JSS	[109]
110	2025	REACCEP: Automated Co-evolution of Production and Test Code Based on Dynamic Validation and Large Language Models	ISSTA	[110]
111	2025	Reflective Unit Test Generation for Precise Type Error Detection with Large Language Models	ASE	[111]
112	2025	Reinforcement Learning from Automatic Feedback for High-Quality Unit Test Generation	DeepTest	[112]
113	2025	Retrieval-Augmented Fine-Tuning for Improving Retrieve-and-Edit Based Assertion Generation	TSE	[113]
114	2025	RUG: Turbo LLM for Rust Unit Test Generation	ICSE	[114]
115	2025	Seed&Steer: Guiding Large Language Models with Compilable Prefix and Branch Signals for Unit Test Generation	arXiv	[115]
116	2025	STRUT: Structured Seed Case Guided Unit Test Generation for C Programs using LLMs	ISSTA	[116]
117	2025	Test Intention Guided LLM-Based Unit Test Generation	ICSE	[117]
118	2025	Test Wars: A Comparative Study of SBST, Symbolic Execution, and LLM-Based Approaches to Unit Test Generation	ICST	[118]
119	2025	TestEval: Benchmarking Large Language Models for Test Case Generation	NAACL	[119]
120	2025	TestForge: Feedback-Driven, Agentic Test Suite Generation	arXiv	[120]
121	2025	TestGenEval: A Real World Unit Test Generation and Test Completion Benchmark	ICLR	[121]
122	2025	TestLoter: A logic-driven framework for automated unit test generation and error repair using large language models	JCL	[122]
123	2025	The Prompt Alchemist: Automated LLM-Tailored Prompt Optimization for Test Case Generation	arXiv	[123]
124	2025	TOGLL: Correct and Strong Test Oracle Generation with LLMs	ICSE	[124]
125	2025	Tracets4J: A Traceable Unit Test Generation Dataset	SANER	[125]
126	2025	Tratto: A Neuro-Symbolic Approach to Deriving Axiomatic Test Oracles	ISSTA	[126]
127	2025	Unify and Triumph: Polyglot, Diverse, and Self-Consistent Generation of Unit Tests with LLMs	arXiv	[127]
128	2025	Unit Test Update through LLM-Driven Context Collection and Error-Type-Aware Refinement	ASE	[128]
129	2025	UTFix: Change Aware Unit Test Repairing using LLM	OOPSLA	[129]
130	2025	Vulnerability-Triggering Test Case Generation from Third-Party Libraries	FORGE	[130]

Continued on next page

ID	Year	Title	Venue	Reference
131	2025	What You See is What You Get: Attention-Based Self-Guided Automatic Unit Test Generation	ICSE	[131]
132	2025	YATE: The Role of Test Repair in LLM-Based Unit Test Generation	arXiv	[132]
133	2026	Advancing Code Coverage: Incorporating Program Analysis with Large Language Models	TOSEM	[133]
134	2026	Agent-Based Test Assertion Generation via Diverse Perspective Aggregation	arXiv	[134]
135	2026	AssertFlip: Reproducing Bugs via Inversion of LLM-Generated Passing Tests	ICSE	[135]
136	2026	ATGen: Adversarial Reinforcement Learning for Test Case Generation	ICLR	[136]
137	2026	Auditing and Decomposing Feedback-Driven Evolution in LLM Test Generation under the Oracle Problem	arXiv	[137]
138	2026	Automated Knowledge-Aware Test Reuse	FSE	[138]
139	2026	Automated Test Suite Enhancement Using Large Language Models with Few-shot Prompting	ICPC	[139]
140	2026	Automated Unit Test Generation via Chain-of-Thought Prompt and Reinforcement Learning from Coverage Feedback	TOSEM	[140]
141	2026	Benchmarking LLMs for Unit Test Generation from Real-World Functions	TOSEM	[141]
142	2026	Beyond Test Presence: Assessing the Quality and Robustness of Agent-Generated Tests in Open-Source Projects	COMPSAC	[142]
143	2026	Boosting unit test generation via structure-aware fine-tuning of pre-trained model	IST	[143]
144	2026	BreakGuard: Towards Detecting Dependency Breaking Changes with LLM-Generated Tests	arXiv	[144]
145	2026	Call-Chain-Aware LLM-Based Test Generation for Java Projects	arXiv	[145]
146	2026	Can Old Tests Do New Tricks for Resolving SWE Issues?	FSE	[146]
147	2026	Can test cases generated by large language models facilitate automated program repair?	EMSE	[147]
148	2026	CASCADE: Detecting Inconsistencies between Code and Documentation with Automatic Test Generation	FSE	[148]
149	2026	CGTest: A hybrid coverage-guided framework for cost-effective unit test generation	JSS	[149]
150	2026	Change And Cover: Last-Mile, Pull Request-Based Regression Test Augmentation	ICSE	[150]
151	2026	CITYWALK: Enhancing LLM-Based C++ Unit Test Generation via Project-Dependency Awareness and Language-Specific Knowledge	TOSEM	[151]
152	2026	CogPath: A Constraint-Guided Context-Reduction Framework for LLM-Based Test Generation	SCAM	[152]
153	2026	Consistency Meets Verification: Enhancing Test Generation Quality in Large Language Models Without Ground-Truth Solutions	ICST	[153]
154	2026	Context Matters: Improving the Practical Reliability of LLM-Based Unit Test Generation (Experience Paper)	ISSTA	[154]
155	2026	Do Coverage and Mutation Scores of LLM-Generated Test Suites Correlate with Their Effectiveness? (Replicability Study)	ISSTA	[155]
156	2026	Documentation vs. Code Patterns: What Drives LLM-Based Exception Oracle Generation?	ASE	[156]
157	2026	DPIAgent: Divide, Protocol, Isolate for Agentic Reproduction Test Generation	arXiv	[157]
158	2026	E-Test: E'er-Improving Test Suites	ICSE	[158]
159	2026	Enabling Global, Human-Centered Explanations for LLMs: From Tokens to Interpretable Code and Test Generation	ICSE	[159]
160	2026	Escaping the Self-Repair Trap: Improving Test Oracle Generation via Dual-Context Awareness	ASE	[160]

Continued on next page

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161	2026	Evaluating and Mitigating the Misguidance Effect of Buggy Code in LLM-Generated Unit Tests	ISSTA	[161]
162	2026	Exploring the integration of large language models in industrial test maintenance processes	JSS	[162]
163	2026	Fail-Aware and Explainable Test Oracle Prediction	arXiv	[163]
164	2026	From Business Requirements to Test Assertions: Evaluating LLM-Generated Oracles on Real Bugs	arXiv	[164]
165	2026	From Greedy Steps to Global Optimization: Learning Sequential Test Suite Generation	ISSTA	[165]
166	2026	Fusing LLMs and Genetic Algorithm for High-Quality Unit Test Generation	TOSEM	[166]
167	2026	Generalizing Test Cases for Comprehensive Test Scenario Coverage	FSE	[167]
168	2026	Generating Project-Specific Test Cases with Requirement Validation Intention	ISSTA	[168]
169	2026	Hallucination to Consensus: Multi-Agent LLMs for End-to-End JUnit Test Generation	TOSEM	[169]
170	2026	Heterogeneous Prompting and Execution Feedback for SWE Issue Test Generation and Selection	ICSE	[170]
171	2026	HieraTest: Hierarchical Dependency-Driven Framework with Multi-Strategy Repair for LLM-Based Unit Test Generation	SANER	[171]
172	2026	How Can ChatGPT Support Human Security Testers to Help Mitigate Supply Chain Attacks?	TSE	[172]
173	2026	How well LLM-based test generation techniques perform with newer LLM versions?	ICST	[173]
174	2026	Humanizing Automatically Generated Unit Test Suites with LLM-Based Refactoring	arXiv	[174]
175	2026	iCoRe: An Iterative Correlation-Aware Retriever for Bug Reproduction Test Generation	FSE	[175]
176	2026	Impact of code context and prompting strategies on automated unit test generation with modern general-purpose large language models	JSS	[176]
177	2026	Improving LLM-Based Unit Test Generation Through Root-Cause-Driven Prompt Design	AITest	[177]
178	2026	IntentTester: Intent-Driven Multi-agent Framework for Cross-Library Test Migration	FSE	[178]
179	2026	Issue2Test: Generating Reproducing Test Cases from Issue Reports	ICSE	[179]
180	2026	JSTestCraft: Addressing Context Deficits in JavaScript Unit Test Generation via Agentic Multi-Level Contextual Analysis	TOSEM	[180]
181	2026	Knowledge Matters: Injecting Project and Testing Knowledge into LLM-based Unit Test Generation	ICSE	[181]
182	2026	Knowledge-Guided Synthetic Bug Feedback for LLM-Based Unit Test Generation	arXiv	[182]
183	2026	Learning to Generate Unit Test via Adversarial Reinforcement Learning	ICLR	[183]
184	2026	Less Is More: Failing Test Generation with Large Language Models	TOSEM	[184]
185	2026	Leveraging Change Types and Contexts to Guide LLMs for Automated Test Code Updating	ICPC	[185]
186	2026	Library-Aware Doubles and Iterative Repair for Large Language Model-Generated Unit Tests in OpenSIL Firmware	arXiv	[186]
187	2026	LLM Test Generation via Iterative Hybrid Program Analysis	ICSE	[187]
188	2026	LLM vs. Human Unit Tests: Fault Detection on Real Python Bugs	arXiv	[188]
189	2026	LLM-Based Detection and Test Generation for Command Injection Vulnerabilities in Python	AITest	[189]
190	2026	LLM-Based Invariant Testing for Software Functional Bugs	ISSRE	[190]

Continued on next page

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191	2026	LLMutantKiller: Using Large Language Models to Generate Tests that Kill Mutants	ISSTA	[191]
192	2026	LSPRAG: LSP-Guided RAG for Language-Agnostic Real-Time Unit Test Generation	ICSE	[192]
193	2026	MACO: Multi-agent collaborative optimization for unit test case generation	IST	[193]
194	2026	Measuring the Influence of Incorrect Code on Test Generation	ICSE	[194]
195	2026	MR-Coupler: Automated Metamorphic Test Generation via Functional Coupling Analysis	FSE	[195]
196	2026	Multi-Agent LLM Collaboration for Unit Test Generation via Human-Testing-Inspired Workflows	arXiv	[196]
197	2026	MultiFileTest: A Multi-File-Level LLM Unit Test Generation Benchmark and Impact of Error Fixing Mechanisms	ACL	[197]
198	2026	MuMuTestUp: Mutation-based Multi-Agent Test Case Update	ISSTA	[198]
199	2026	Mutation-Guided Unit Test Generation With a Large Language Model	TSE	[199]
200	2026	Ockhamareto: Pareto-Gated Segment-Level Credit Assignment for Concise Unit-Test Generation with Reinforcement Learning	arXiv	[200]
201	2026	On the Diffusion of Test Smells in LLM-Generated Unit Tests	TOSEM	[201]
202	2026	On the risk of coding before testing: An empirical study on LLM-based test generation workflow	arXiv	[202]
203	2026	Optimizing Context and Cost in LLM-Based Unit Test Generation: A Study on External Dependency Retrieval Strategies	Expert Systems	[203]
204	2026	PALM: Path-aware LLM-based Test Generation with Comprehension	ICPC	[204]
205	2026	Planning to Explore: Curiosity-Driven Planning for LLM Test Generation	arXiv	[205]
206	2026	PR-Aware Automated Unit Test Generation: Challenges and Opportunities	COMPSAC	[206]
207	2026	Probe to Generate: Program Variant-Guided Test Augmentation for Repository-Level Repair Benchmarks	ASE	[207]
208	2026	Prompt engineering in LLMs for automated unit test generation: A large-scale study	EMSE	[208]
209	2026	Reference-Based Retrieval-Augmented Unit Test Generation	TOSEM	[209]
210	2026	ReProAgent: Tool-Augmented Multi-Stage Agentic Generation of Bug Reproduction Tests from Issue Reports	arXiv	[210]
211	2026	Retrieval-Augmented Test Generation: How Far Are We?	ICSE	[211]
212	2026	Retrieval-Augmented Unit Test Suggestion Generation	TOSEM	[212]
213	2026	RM-RF: Reward Model for Run-Free Unit Test Evaluation	SANER	[213]
214	2026	SCATE: Learning to Supervise Coding Agents for Cost-Effective Test Generation	arXiv	[214]
215	2026	Synthesizing File-Level Data for Unit Test Generation with Chain-of-Thoughts via Self-Debugging	ASE	[215]
216	2026	TATG: Tracking-Aware Testing Objective for LLM-based Test Generation	arXiv	[216]
217	2026	Test vs Mutant: Adversarial LLM Agents for Robust Unit Test Generation	ISSTA	[217]
218	2026	TestBench: Evaluating Class-Level Test Case Generation Capability of Large Language Models	FCS	[218]
219	2026	TestEvo-Bench: An Executable and Live Benchmark for Test and Code Co-Evolution	arXiv	[219]
220	2026	TestForge: Benchmarking LLM-Based Test Case Generation	SANER	[220]
221	2026	Testora: Using Natural Language Intent to Detect Behavioral Regressions	ICSE	[221]
222	2026	TestTailor: Generating High-Coverage Tests via Path-Proximal Tests with LLMs	FSE	[222]

Continued on next page

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223	2026	TestWeaver: Execution-aware, Feedback-driven Regression Testing Generation with Large Language Models	ICSE	[223]
224	2026	The Quality of Claude AI-authored Python Tests Is Not Weaker Than Human-authored Tests	arXiv	[224]
225	2026	Toward Automated Validation of Language Model Synthesized Test Cases Using Semantic Entropy	TSE	[225]
226	2026	Type-aware LLM-based Test Generation for Python Programs	TOSEM	[226]
227	2026	Uncovering Business Logic Bugs via Semantics-Driven Unit Test Generation (Experience Paper)	ISSTA	[227]
228	2026	XREPOTEST: Benchmarking Multilingual Repository-Level Unit Test Generation for Large Language Models	arXiv	[228]

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- [3] Elizabeth Dinella, Gabriel Ryan, Todd Mytkowicz, and Shuvendu K. Lahiri. TOGA: A neural method for test oracle generation. In *44th IEEE/ACM 44th International Conference on Software Engineering, ICSE 2022, Pittsburgh, PA, USA, May 25-27, 2022*, pages 2130–2141. ACM, 2022.
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- [9] Xing Hu, Zhuang Liu, Xin Xia, Zhongxin Liu, Tongtong Xu, and Xiaohu Yang. Identify and update test cases when production code changes: A transformer-based approach. In *38th IEEE/ACM International Conference on Automated Software Engineering, ASE 2023, Luxembourg, September 11-15, 2023*, pages 1111–1122. IEEE, 2023.
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